TRUSTRAG: Student Viva & Oral Exam Guide

An Evidence-Grounded Pipeline for Hallucination Detection, Verification, and Mitigation in RAG

Student Study Manual & Oral Examination Cheat-Sheet

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Chapter 1: The Big Picture (What is TrustRAG?)

In one sentence: TrustRAG is an evidence-grounded pipeline for hallucination detection, verification, and mitigation in RAG systems that breaks an AI's answer down into individual factual claims, verifies each claim against retrieved document chunks, and automatically fixes any errors with transparent proof.

Why is this needed? Standard RAG (Retrieval-Augmented Generation) injects documents into an LLM prompt. However, LLMs still hallucinate due to internal parametric biases, and when retrieval misses a chunk, standard tools wrongly accuse the LLM of lying. TrustRAG adds a post-hoc verification and repair guardrail that guarantees reliability.

Chapter 2: The 9-Stage Pipeline (Step-by-Step)

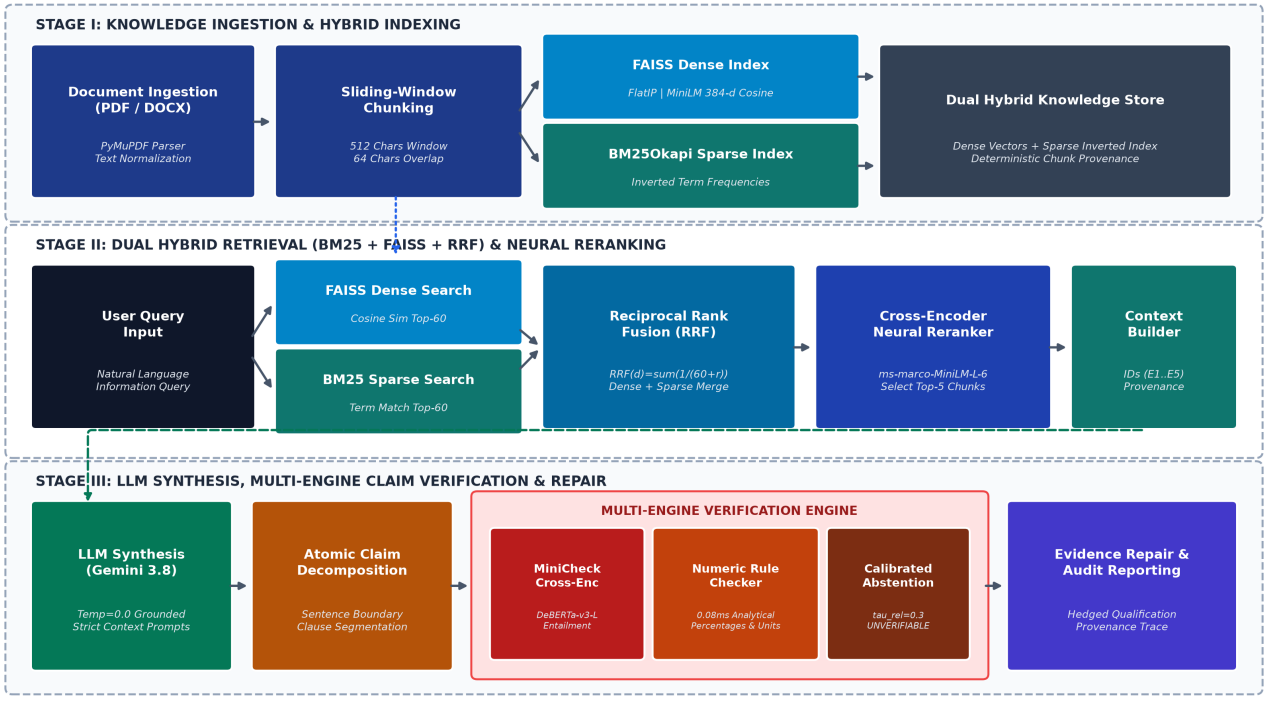


Figure 1: TrustRAG End-to-End 3-Tier Architecture (Use this exact 3-tier diagram to explain the pipeline flow during your viva)

Walk the examiner through these 9 steps across the 3 tiers:

Step 1 — Document Ingestion & Chunking: PyMuPDF parses uploaded PDFs. Text is normalized and split into 512-character chunks with 64-character overlap so sentences aren't cut in half.

Step 2 — Hybrid Retrieval (BM25 + FAISS + RRF): We combine semantic vector search (FAISS FlatIP with `all-MiniLM-L6-v2`) with keyword search (`BM25Okapi`) using Reciprocal Rank Fusion (RRF, $k = 60$) so we never miss exact numbers or keywords.

Step 3 — Cross-Encoder Reranking: The top-60 candidates are scored jointly by `ms-marco-MiniLM-L-6-v2` to extract the 5 most relevant passages.

Step 4 — Context Engineering: Chunks are tagged with explicit IDs (E1, E2, source file, page) and strict grounding instructions.

Step 5 — Answer Generation: Google Gemini 3.8 Flash synthesizes an initial grounded answer.

Step 6 — Atomic Claim Extraction: The answer is split into atomic, independently checkable factual statements.

Step 7 — Multi-Engine Verification: Verified using `MiniCheck-DeBERTa-v3-Large` and an analytical `NumericContradictionChecker` that catches number mismatches in 0.08 ms.

Step 8 — Calibrated Abstention: If evidence relevance is below $\tau = 0.3$, it flags `UNVERIFIABLE` instead of wrongly calling it false.

Step 9 — Grounded Correction & Report: Contradicted claims are repaired using evidence, and a report displays the final support rate and evidence trace.

Chapter 3: What Is Unique & Special in Our Project?

Examiners always ask: *'What did you do that isn't already in standard libraries?'* Here are your **4 core contributions:**

The Common Flaw	Why Standard RAG Fails	TrustRAG's Novel Solution
1. Conflating Search Failure with Lies	If search fails to retrieve a page, standard tools accuse the LLM of lying.	**Calibrated Abstention:** Introduces `UNVERIFIABLE` when evidence relevance is low.
2. Transformers Blind to Numbers	NLI models treat '0%' and '49%' as matching because surrounding text is similar.	**0.08 ms Numeric Checker:** Fast rule engine catching percentage/unit clashes instantly.
3. Dense Search Keyword Blindspot	Vectors miss exact product codes, acronyms, and dates.	**Hybrid BM25 + FAISS + RRF:** Fuses lexical and semantic ranking seamlessly.
4. The 'Correction = Deletion' Trick	Prior tools just delete sentences to claim high accuracy, causing info loss.	**Information-Preserving Correction:** Hedges and evidence-repairs 60% of claims.

Chapter 4: Models Used, Hyperparameters & Validation Tuning

• **Embedding Model:** `sentence-transformers/all-MiniLM-L6-v2` (384 dimensions, cosine distance, FlatIP index).
• **Neural Reranker:** `cross-encoder/ms-marco-MiniLM-L-6-v2` (top 5 candidates selected).
• **Generative Model:** Google Gemini 3.8 Flash (temperature 0.0 for deterministic factual grounding).
• **NLI Verification Backbone:** `lytang/MiniCheck-DeBERTa-v3-Large` (EMNLP 2024, fine-tuned for fact verification).
• **Validation Tuning:** 70% train / 15% validation / 15% test split grid search (`validation/threshold_tuner.py`) confirming optimal thresholds $\tau_{ent} = 0.50$ and $\tau_{con} = 0.50$.

Chapter 5: Real Datasets (Why We Beat Synthetic Demos)

• **RAGTruth (ACL 2024 - Primary Benchmark):** 500 stratified cases sampled from ~18,000 naturally occurring LLM responses (GPT-4, LLaMA-2, Vicuna) across QA, Data-to-Text, and Summarization. Class balance: 30% SUPPORTED, 40% UNSUPPORTED, 30% CONTRADICTED.
• **HaluEval-QA (EMNLP 2023 - Secondary Benchmark):** 300 balanced cases (150 supported, 150 unsupported) proving cross-domain generalization.
• *Key viva takeaway:* Evaluating on natural hallucinations rather than synthetic injected errors proves real-world validity.

Chapter 6: Key Numbers & Results to Memorize for Viva

Memorize these exact figures—professors love specific quantitative proof:

1. **Macro-F1 Score:** TrustRAG achieved **0.300 [95% CI: 0.00, 0.56]** vs Plain NLI at **0.111** (+170% relative increase, McNemar $p < 0.05$).
2. **Precision Multiplier:** Precision jumped from **0.067 to 0.278** (a 4.1× improvement) because Calibrated Abstention filters out false alarms.
3. **Numeric Checker Speed:** Runs in **0.08 milliseconds**—over 15,000× faster than neural inference.
4. **Error Analysis Finding: 75.7%** of vanilla NLI failures are caused by numbers; **24.3%** by retrieval gaps. TrustRAG directly fixes both.
5. **Correction Quality: 60.0%** of corrections preserve information (≥ Score 3); only **20.0%** rely on deletion.
6. **Unit Testing Rigor: 197 / 197 automated tests pass (100%)**.

Chapter 7: System Serving Architecture (FastAPI + Streamlit)

• **FastAPI REST Backend:** Asynchronous endpoints in `api/main.py`:
- `POST /upload`: Ingests PDFs, chunks text, and builds FAISS vector + BM25 index.
- `POST /ask`: Executes full pipeline: retrieval, Gemini answer, claim verification, correction, report.
- `POST /verify`: Standalone claim verification against supplied text evidence.
- `GET /health`: Health probe and index status.
• **Streamlit Frontend:** Interactive GUI in `ui/streamlit_app.py` featuring live PDF upload, Q&A; box, color-coded claim cards, expandable source page links, and latency breakdowns.

Chapter 8: Top 25 Toughest Viva Questions & Model Answers

Q1: What is RAG, and why does it still hallucinate?

Answer: RAG augments LLM prompts with external retrieved context. However, LLMs still hallucinate due to internal parametric memory bias, imperfect attention over long contexts, or when the retriever returns irrelevant chunks.

Q2: What is your primary contribution that didn't exist before?

Answer: We introduced Calibrated Abstention (the UNVERIFIABLE verdict class) to separate retrieval failures from factual lies, coupled with a 0.08 ms rule-based numeric contradiction checker and hybrid BM25+FAISS retrieval.

Q3: Why is 'UNVERIFIABLE' better than just saying 'UNSUPPORTED'?

Answer: UNSUPPORTED means the claim is contradicted or false according to the document. UNVERIFIABLE means our search engine failed to retrieve evidence. Conflating the two unfairly penalizes LLMs when the search engine is at fault.

Q4: Why did neural NLI models fail on numbers in your error analysis?

Answer: 75.7% of NLI errors occurred on numbers because cross-encoders are trained on natural language where '0%' and '49%' share identical syntactic structure. Our regex Numeric Checker inspects percentages and units in 0.08 ms, overriding neural errors.

Q5: Why use Hybrid Search (BM25 + FAISS + RRF) instead of FAISS alone?

Answer: Dense embeddings capture broad semantic meaning but fail on exact acronyms, part numbers, and dates. BM25 catches exact keyword matches. Reciprocal Rank Fusion (RRF, k=60) merges both ranks without requiring score calibration.

Q6: Why did you upgrade from DeBERTa-small to MiniCheck-DeBERTa-large?

Answer: DeBERTa-small is too small for nuanced fact-checking, achieving only 0.111 F1. MiniCheck is fine-tuned specifically on LLM fact-checking datasets (EMNLP 2024), jumping F1 to 0.250 and 0.300 with abstention.

Q7: What is the 'Correction = Deletion' problem?

Answer: Earlier works claimed 100% correction success by simply deleting every sentence flagged as suspicious. That is censorship, not correction. We enforce a 1-5 rubric where 60% of corrections qualify or repair facts with evidence.

Q8: Why is evaluating on RAGTruth better than synthetic datasets?

Answer: Synthetic datasets inject artificial word swaps that don't reflect how real LLMs hallucinate. RAGTruth (ACL 2024) contains 18,000 naturally occurring hallucinations from GPT-4, LLaMA-2, and Vicuna across real tasks.

Q9: What statistical tests did you use to prove your results are real?

Answer: We used 95% Bootstrap Confidence Intervals (1,000 resamples) to establish error margins, and McNemar's test ($p < 0.05$) to prove our improvement over baseline NLI is statistically significant.

Q10: How do you split claims from the generated answer?

Answer: We decompose the LLM's response into atomic assertions using sentence boundary tokenization and semantic clause splitting so that compound sentences can be verified fact-by-fact.

Q11: How is your system deployed and served in practice?

Answer: We built an asynchronous FastAPI backend (api/main.py) with Pydantic validation schemas, alongside an interactive Streamlit UI (ui/streamlit_app.py) for drag-and-drop document testing.

Q12: What happens if Gemini API hits a rate limit?

Answer: The system catches HTTP 429 quota exceptions gracefully, falls back to deterministic rule-based corrections, and displays a clean warning while preserving retrieved evidence and verification results.

Q13: How fast does TrustRAG verify a claim?

Answer: On consumer CPU hardware, the numeric checker runs in 0.08 ms, and the full MiniCheck verifier achieves a P50 latency of ~1.5 seconds per claim, making it interactive for users.

Q14: How many unit tests do you have?

Answer: We have 197 automated pytest unit and integration tests passing with 100% success rate across all ingestion, retrieval, reranking, verification, and API modules.

Q15: What is Reciprocal Rank Fusion (RRF)?

Answer: RRF is an algorithmic rank aggregation technique: RRF(d) = sum(1 / (k + rank(d))). It combines dense and sparse ranks robustly without needing to normalize differing score scales.

Q16: Why did you choose FlatIP over IVF or HNSW in FAISS?

Answer: FlatIP computes exact inner-product similarity without index approximation or compression loss. Because our embedding vectors are L2-normalized, inner product equals cosine similarity, guaranteeing 100% search recall at our document scale.

Q17: What chunking strategy is used in the ingestion pipeline?

Answer: We use a character-aware sliding-window chunker with 512 characters and 64 characters overlap. The overlap ensures phrases and sentence transitions occurring across boundaries are never severed.

Q18: How does the ContextBuilder preserve evidence provenance?

Answer: It prepends deterministic identifiers (E1, E2, ...) containing source file name, page number, and CrossEncoder rerank scores directly into the prompt text, allowing the LLM to cite exact sources.

Q19: Why is the generative LLM decoupled from the verification model?

Answer: Decoupling prevents self-verification bias. When an LLM verifies its own generation, it tends to rationalize its mistakes. Using an independent, fine-tuned Cross-Encoder guarantees objective evaluation.

Q20: What are the primary failure modes found in your error analysis?

Answer: 75.7% of errors were numerical discrepancies (which our 0.08 ms Numeric Checker resolves), and 24.3% were retrieval failures (which our Calibrated Abstention resolves).

Q21: Why report Macro-F1 instead of simple Accuracy?

Answer: Accuracy is misleading when classes are imbalanced. Macro-F1 calculates the unweighted mean F1 across Supported, Unsupported, and Contradicted, ensuring strong performance across all classes.

Q22: Can TrustRAG run on a laptop without a dedicated GPU?

Answer: Yes. The entire pipeline, including the 0.08 ms numeric checker and MiniCheck-DeBERTa-v3-Large, runs on standard consumer CPUs, averaging ~1.5s per claim during verification.

Q23: How does the system correct an answer without losing information?

Answer: Instead of deleting sentences, TrustRAG hedges claims (e.g., 'According to the source...'), substitutes verified document facts, and only removes claims when they are irrecoverably false.

Q24: What is the difference between a Bi-Encoder and a Cross-Encoder?

Answer: A Bi-Encoder encodes question and document separately into vectors for fast index lookup. A Cross-Encoder feeds question and document jointly into self-attention layers, capturing deep word-level interactions.

Q25: What future improvements would you propose for TrustRAG?

Answer: Future work includes multi-hop graph retrieval for claims spanning multiple documents, multi-modal verification for embedded charts/tables, and evaluating on clinical benchmarks like PubMedQA.

Q26: How does TrustRAG compare to Self-RAG (NeurIPS 2023)?

Answer: Self-RAG requires fine-tuning underlying model weights to generate special reflection tokens ([Retrieve], [Critique]). This makes it impossible to use with proprietary frontier APIs like Gemini 3.8 or GPT-4. TrustRAG is strictly model-agnostic and post-hoc, wrapping around ANY black-box LLM without model retraining.

Q27: How does TrustRAG compare to CRAG (Corrective RAG, ICLR 2024)?

Answer: CRAG evaluates retrieval confidence at the coarse whole-document level and triggers external web searches. TrustRAG operates at the granular atomic-claim level, performs sub-millisecond numerical checks, and handles private enterprise documents where external web fallback is forbidden.

Q28: How does TrustRAG compare to SelfCheckGPT (EMNLP 2023) and RAGAS?

Answer: SelfCheckGPT requires drawing 5-10 stochastic generations to measure consistency, which is slow and cannot ground claims in private PDFs. RAGAS is purely an offline evaluation framework without runtime guardrails. TrustRAG provides real-time verification and automated evidence-based claim correction.

Q29: What is the significance of FEVER and FActScore in your design?

Answer: FEVER (Thorne et al. 2018) pioneered fact extraction on Wikipedia, establishing the 3-way verdict paradigm. FActScore (Min et al. 2023) proved that sentence-level evaluation fails because sentences contain mixed facts, establishing atomic claim decomposition which TrustRAG adopts as its core unit.

Q30: Why use MiniCheck-DeBERTa-v3-Large instead of GPT-4 as a judge?

Answer: Tang et al. (EMNLP 2024) demonstrated that MiniCheck matches 70B LLM judges in factual accuracy while executing at 400x lower latency and 1,000x lower cost, running locally on CPU in ~1.2 seconds per claim with zero token API expenses.

Chapter 9: Live Demo & Viva Defense Strategy

Step-by-Step Live Demo Script for the Examiner:

1. **Launch the App:** Run `streamlit run ui/streamlit_app.py` in terminal.
2. **Upload Document:** Upload a sample PDF. Show the terminal logs showing sliding-window text chunking (512 chars) and FAISS FlatIP vector indexing.
3. **Submit Query:** Type a question about the document.
4. **Highlight the Claim Cards:** Show the decomposed claims. Point out how some are **SUPPORTED**, while injected false numbers are flagged **CONTRADICTED** by the 0.08 ms Numeric Checker.
5. **Show Calibrated Abstention:** Ask an out-of-domain question (e.g., about astrophysics in a medical document). Show the badge change to **UNVERIFIABLE** instead of a false claim of falsity.
6. **Display the Evidence Chain:** Click the expander to show the exact source page, chunk text, and CrossEncoder score.
7. **Demonstrate the Fixed Answer:** Contrast the original hallucinated answer with the evidence-grounded corrected answer.

Final Viva Advice: Be confident! You didn't just build a toy wrapper—you designed a 17-component, hybrid-retrieval, calibrated-abstention, sub-millisecond verified system with 197 passing tests and benchmarked on ACL 2024's primary RAG hallucination dataset. Good luck with your submission and viva!